

# Siddhartha Shakya

Portfolio | LinkedIn | GitHub | siddhuushakya@gmail.com | Dubai, UAE | +971 55 183 0660

## PROFILE SUMMARY

Quick learner and highly efficient **Computer Science graduate** currently pursuing **PG Diploma in Data Science** (evening batch) at MAHE Dubai. Proficient in **Python**, **SQL**, and **Excel** with hands-on experience in data science projects revolving around data **analytics**, **engineering** and **AI/ML**. Proven ability to deliver data-driven solutions that create measurable business impact. Valid UAE work authorization on **student visa**.

## TECHNICAL SKILLS

|                           |                                                                      |
|---------------------------|----------------------------------------------------------------------|
| <b>Languages</b>          | Python, SQL, C/C++                                                   |
| <b>Databases</b>          | PostgreSQL, MySQL, MongoDB, SQLite                                   |
| <b>Data Analytics</b>     | Pandas, Matplotlib, Seaborn, Streamlit, Excel                        |
| <b>Data Engineering</b>   | Databricks, Apache Airflow, Snowflake, dbt, PySpark                  |
| <b>AI &amp; ML</b>        | PyTorch, Scikit-learn, NumPy, Jupyter Notebooks, Google Colab        |
| <b>DevOps &amp; Tools</b> | Docker, Git/GitHub, GitHub Actions (CI/CD), REST APIs, Linux, MLflow |

## WORK EXPERIENCE

**AI Intern** | *Mach24 Orbitals* **Oct 2025 – Feb 2026**

- Designed and implemented an automated geospatial data extraction system leveraging GEE (Google Earth Engine) API to retrieve **1,000+** satellite imagery datasets monthly across custom AOIs and timeframes, eliminating manual download processes and reducing data acquisition time by **75%**.
- Owned end-to-end data pipeline responsibility: processed **1,000+** monthly datasets through automated ingestion, quality assurance, and normalization, achieving **83%** data accuracy and reducing manual preprocessing time by **60%**.

**Graduate Teaching Assistant** | *Herald College Kathmandu* **Aug 2024 – Sep 2025**

- Developed comprehensive Big Data curriculum covering Hadoop ecosystem, SQL and MongoDB databases, Python, and PySpark; delivered hands-on labs and practical assignments to **120+** students, achieving **15%** improvement in assessment scores.
- Architected and delivered High Performance Computing (HPC) workshops on parallel programming, C/C++, and CUDA GPU computing; built a password-cracking lab showcasing **6x speedup** via multithreading and **40x speedup** via CUDA over a single-core baseline.

## PROJECTS

**EduFin Investigation** | *Financial Data Analytics* **2026**

- Built and optimized data pipelines in Databricks using PySpark and SQL to ingest and analyze a 15,000+ student loan portfolio, uncovering a Rs. 12 Crore financial discrepancy.
- Identified critical risk patterns through segmentation analysis, isolating high-risk regions, customer profiles, and partner institutions to inform data-driven risk mitigation strategies.
- **Tech Stack:** Python, Databricks, PySpark SQL, Delta Tables, Data Pipelines

**NYC Parking Violations** | *Data Engineering* **2025**

- Built a multi-layer Medallion Architecture (Bronze/Silver/Gold) ELT pipeline transforming raw NYC parking violation data from NYC Open Data into analytics-ready metric tables.
- Automated production deployments using GitHub Actions with separate dev/prod DuckDB profiles.
- **Tech Stack:** dbt Core, DuckDB, SQL, Python, Jupyter Notebook, GitHub Actions, YAML

## EDUCATION

**PG Diploma in Data Science** **Feb 2026 – present**  
– *Manipal Academy of Higher Education*

**BSc (Hons) in Computer Science** **Feb 2022 – Jun 2024**  
– *University of Wolverhampton*  
– Honors: **First Class**

## CERTIFICATIONS

|                                                |           |             |
|------------------------------------------------|-----------|-------------|
| <b>AI Fluency: Framework &amp; Foundations</b> | Anthropic | <b>2026</b> |
| <b>MLOps Bootcamp</b>                          | Udemy     | <b>2025</b> |
| <b>SQL Bootcamp</b>                            | Udemy     | <b>2025</b> |